

Vishal Kumar Allam

Seattle, WA | 816-666-2813 | itsmevishal36@gmail.com | **Work Authorization: H1B**

PROFESSIONAL SUMMARY

- Data Engineering professional with **6+ years of experience** designing, developing, optimizing, and supporting complex enterprise data pipelines and data platforms using Python, SQL, Snowflake, PostgreSQL, MySQL, Spark, Airflow, and cloud technologies.
- Hands-on experience building **batch and streaming data pipelines** that ingest, transform, model, and distribute structured and unstructured data from APIs, databases, files, event streams, and enterprise applications.
- Strong expertise in **complex SQL, data modeling, Snowflake, relational databases, dimensional design, schema evolution, performance tuning, and resilient data structures** that make information accessible, performant, and adaptable to change.
- Experienced designing **data exchange systems and API-driven integrations** across engineering, analytics, and business applications using Python, REST APIs, message queues, event-driven patterns, and stream-processing technologies.
- Built and maintained production data solutions across **AWS, Azure, Docker, Kubernetes, Airflow, Spark, Git, Jenkins, and CI/CD**, supporting reliable deployments across development, staging, and production environments.
- Strong focus on **data accuracy, validation, reconciliation, operational excellence, automation, performance optimization, monitoring, alerting, troubleshooting, and production support** for business-critical data systems.
- Collaborated with **engineering teams, analytics users, product owners, and technical/non-technical stakeholders** to gather requirements, translate business needs into technical designs, resolve ambiguity, and deliver scalable data solutions in fast-paced environments.

EDUCATION

University of Missouri Kansas City

"Master of Science" in Computer Science

Kansas City, MO

Jan. 2023 – May 2024

Jawaharlal Nehru Technological University

"Bachelor of Technology" in Computer Science

Hyderabad, India

Aug. 2015 - May 2019

TECHNICAL SKILLS

- **Programming & Querying:** Python, Java, SQL, Complex SQL, PySpark, Spark SQL, Pandas, Data Processing, Data Transformation, Object-Oriented Programming, API Development.
- **Databases & Data Modeling:** Snowflake, PostgreSQL, MySQL, SQL Server, Oracle, Relational Databases, Data Modeling, Dimensional Modeling, Schema Design, Schema Evolution, Query Optimization.
- **Data Engineering:** ETL/ELT, Batch Pipelines, Streaming Pipelines, Data Ingestion, Structured Data, Unstructured Data, Data Exchange Systems, Data Validation, Reconciliation, Data Quality, Data Integration.
- **Streaming & Orchestration:** Apache Airflow, Apache Spark, Kafka, Message Queues, Stream Processing, Event-Driven Architecture, Workflow Orchestration, Scheduled Pipelines, Distributed Processing.
- **Cloud & Infrastructure:** AWS, Azure, GCP Concepts, S3, Lambda, Cloud Storage, Docker, Kubernetes, Terraform, Cloud-Native Architecture, Scalable Data Platforms.
- **DevOps & Operations:** Git, Jenkins, Azure DevOps, GitLab CI/CD, CI/CD Pipelines, Environment Management, Monitoring, Logging, Alerting, Production Support, On-Call Support, Incident Resolution.
- **Engineering & Delivery:** Performance Optimization, Process Automation, Coding Best Practices, Technical Requirements, Stakeholder Communication, Agile/Scrum, Root-Cause Analysis, Operational Excellence, Documentation.

Test Automation Engineer

PCMI LLC (Full Time)

Oct 2024 – Present

Seattle, WA, USA

- Designed, developed, and supported complex **enterprise data pipelines** using Python, SQL, PySpark, Palantir Foundry, Snowflake, and cloud services, integrating insurance and operational data from databases, APIs, files, and enterprise applications.
- Built scalable **batch and streaming pipelines** using Spark, Kafka-style event processing, scheduled orchestration, and distributed data-processing patterns to deliver reliable datasets for analytics, reporting, and downstream applications.
- Developed advanced **SQL transformations and data models** across Snowflake, PostgreSQL, and relational platforms, optimizing schemas, joins, aggregations, partition strategies, and query execution for accessibility and performance.
- Designed reusable **data exchange and API integration services** using Python, FastAPI, REST APIs, event-driven workflows, and structured/unstructured data formats to enable reliable communication between source applications and data consumers.
- Implemented strong **data accuracy, reconciliation, validation, duplicate detection, completeness checks, and source-to-target controls**, ensuring production data remained consistent and dependable across business-critical workflows.
- Improved operational excellence through **pipeline automation, performance tuning, Git, CI/CD, Docker, Kubernetes, monitoring, logging, alerting, and environment management** across development and production platforms.
- Partnered with engineering teams, analytics users, product owners, and business stakeholders to **gather requirements, design scalable solutions, resolve production issues, communicate dependencies, and support business-critical data products through ongoing operational support**.

Environments: Python, SQL, PySpark, Apache Spark, Snowflake, Palantir Foundry, PostgreSQL, REST APIs, FastAPI, Batch Pipelines, Streaming Pipelines, Kafka Concepts, Structured/Unstructured Data, AWS, Airflow Concepts, Docker, Kubernetes, Git, CI/CD, Monitoring, Alerting, Production Support.

Software Engineer

Dec 2023 – Sept 2024

Client: CVS

Seattle, WA, USA

- Designed and developed scalable **healthcare data pipelines and backend data services** using Python, SQL, AWS, Spark, and relational databases, supporting analytics and operational use cases across enterprise healthcare systems.
- Built production **batch and near-real-time ingestion pipelines** that processed structured and unstructured data from REST APIs, relational databases, files, and cloud storage into standardized downstream datasets.
- Developed complex **SQL transformations and data models** using PostgreSQL, MySQL, and cloud data technologies, improving data organization, query performance, downstream accessibility, and resilience to source-system changes.
- Implemented distributed processing and integration workflows using **Python, PySpark, Spark, REST APIs, event-driven patterns, and message-processing concepts**, enabling reliable data exchange between enterprise applications.
- Applied rigorous **data validation, reconciliation, completeness, consistency, and accuracy controls** across high-volume healthcare datasets while investigating anomalies and correcting upstream/downstream processing issues.
- Supported cloud and production operations through **AWS, Docker, Kubernetes concepts, Git, Jenkins, CI/CD, CloudWatch, logging, monitoring, alerting, and environment configuration** across development and production workloads.
- Collaborated with engineering, analytics, architecture, and business teams to **translate requirements into technical designs, optimize data workflows, resolve production incidents, automate manual processes, and improve data-platform operational efficiency**.

Environments: Python, SQL, AWS, PySpark, Apache Spark, PostgreSQL, MySQL, REST APIs, Batch Processing, Streaming Concepts, Structured/Unstructured Data, Docker, Kubernetes Concepts, Git, Jenkins, CI/CD, CloudWatch, Monitoring, Logging, Performance Tuning, Production Support.

Software Engineer

Sept 2021 – Dec 2022

Infor

Hyderabad, India

- Designed and developed enterprise **data pipelines and integration solutions** using Python, SQL, PySpark, Spark, and relational databases, supporting operational applications, analytics, and reporting across distributed business systems.

- Built reusable **batch data pipelines and ingestion workflows** to process information from databases, APIs, files, and enterprise platforms, transforming structured and semi-structured data into reliable downstream datasets.
- Developed complex SQL logic and **data models** across PostgreSQL, MySQL, and Oracle, implementing joins, aggregations, stored logic, indexing, schema design, and query optimization for large enterprise workloads.
- Implemented Python and Spark-based processing for **data cleansing, transformation, enrichment, validation, and distributed computation**, improving scalability and reducing processing bottlenecks across data workflows.
- Designed service integrations using **REST APIs, event-driven patterns, asynchronous processing, and reusable data exchange components**, enabling reliable data flow between enterprise applications and analytics platforms.
- Applied engineering practices using **Git, Jenkins, CI/CD, Docker, logging, monitoring, deployment automation, and performance tuning**, improving maintainability and production reliability of data systems.
- Worked with architects, product teams, analysts, and business stakeholders to **gather requirements, translate business needs into technical solutions, troubleshoot data issues, automate manual workflows, and support production data services**.

Environments: Python, SQL, PySpark, Apache Spark, PostgreSQL, MySQL, Oracle, REST APIs, ETL/ELT, Batch Pipelines, Data Modeling, Structured Data, Event-Driven Architecture, Git, Jenkins, CI/CD, Docker, Monitoring, Performance Optimization, Production Support.

Associate Engineer

June 2019 – August 2021

Virtusa

Hyderabad, India

- Developed enterprise **data-processing and integration solutions** using Python, SQL, PostgreSQL, MySQL, and backend technologies, supporting business applications, reporting, and operational data workflows.
- Built reusable **batch ingestion and transformation pipelines** to extract information from databases, files, REST APIs, and enterprise systems and deliver standardized datasets to downstream consumers.
- Developed SQL-based data models and transformation logic using **complex joins, aggregations, schema design, indexing, data normalization, and query optimization**, improving accessibility and performance of operational datasets.
- Implemented Python components for **data cleansing, validation, transformation, reconciliation, scheduling, and API-based integration**, reducing repetitive manual data-processing activities.
- Supported distributed integrations using **REST APIs, event-driven concepts, message-processing patterns, relational databases, and asynchronous workflows**, enabling reliable data exchange across multiple systems.
- Applied software delivery and operational practices using **Git, CI/CD, Docker, Linux, logging, monitoring, troubleshooting, deployment workflows, and production support** across enterprise environments.
- Collaborated with engineers, analysts, architects, and business users to **clarify requirements, translate business needs into technical implementations, troubleshoot data inconsistencies, optimize processes, and maintain reliable production data workflows**.

Environments: Python, SQL, PostgreSQL, MySQL, REST APIs, Batch Pipelines, Data Integration, Data Modeling, Structured/Unstructured Data, Message Processing Concepts, Git, CI/CD, Docker, Linux, Logging, Monitoring, Query Optimization, Production Support.

AWARDS & ACHIEVEMENTS

- 2019-11 Oracle Certified Associate, JAVA SE8 Programmer.
- ISTQB Certified Professional.
- Microsoft Technology Associate (MTA) – Database Administration Fundamentals, HTML- 5 Application Development Fundamentals.
- Awarded “Software Process Achievement” Award for my contribution towards the project in Virtusa.
- Received Customer Appreciation mails for quality releases.